

Conditional Scoring Produces Target-Specific Subsets

The scorer consumes (z_x, z_p, z_t) and predicts $(\hat{\mu}, \hat{\sigma}) = s_\phi(z_x, z_p, z_t)$. Here $\hat{\mu}$ gives predicted mean utility, while $\hat{\sigma}$ gives a positive, input-dependent uncertainty scale. Separate encoders map the three inputs into hidden vectors. FiLM modulation (Perez et al. 2018) uses the PEFT and task codes to rescale and shift sample features. A bilinear term represents sample-PEFT compatibility.

Stage A distills gradient-proxy ranks to learn broad conditional structure. Stage B continues proxy distillation and adds ranking and regression on short-update labels plus a heteroscedastic uncertainty loss. Here *heteroscedastic* means that predicted noise can differ across triples. PCU-Select selects the Stage B checkpoint on held-out PEFT-task pairs. It also fixes the sketch size, label allocation, anchors, horizons, and scalar selection hyperparameters on these pairs before downstream evaluation.

A target’s structural support level determines whether PCU-Select applies the scorer directly or calibrates it. Relative to the seen registry, L0 resembles a seen configuration, L1 makes a larger shift within a seen family, and L2 introduces an unseen family. Seen targets use the trained scorer directly. L0 targets run *zero-shot*: PCU-Select scores a new configuration without target-specific short-update labels. L1 and L2 targets use calibrated scoring when compatible labels exist and a per-target selector otherwise.

After any required calibration, PCU-Select converts the two scorer outputs into a conservative ranking score:

$$q(x) = \hat{\mu}(x, p^*, t^*) - \lambda \hat{\sigma}(x, p^*, t^*), \quad \lambda = 0.2. \quad (8)$$

The uncertainty scale reduces the rank of examples whose predicted utility is noisy. It serves as a ranking penalty, not a confidence interval.

Ranking by q alone can over-select near-duplicates, so PCU-Select clusters examples in semantic space before assigning the budget. It allocates budget B across K clusters C_k , where $k \in \{1, \dots, K\}$:

$$b_k = \text{round} \left(B \frac{(v_k^+)^{\gamma} |C_k|^{1-\gamma}}{\sum_j (v_j^+)^{\gamma} |C_j|^{1-\gamma}} \right), \quad \gamma = 0.6. \quad (9)$$

Here v_k is the mean conservative score among the top 10% of examples in C_k , and $v_k^+ = \max(v_k, 0)$. The factors v_k^+ and $|C_k|$ balance cluster value and size. For the 300K pool, $K = \max(50, \lfloor \sqrt{N} \rfloor)$ produces 547 clusters of about 548 examples. The reported 0.36 GPU-hours per PEFT-task application includes clustering. PCU-Select takes the top- b_k examples by q within each cluster. Non-positive clusters receive no budget; deterministic remainder handling makes the quotas sum to B .

For seen and L0 targets, all gradient extraction and label construction occur offline; online selection uses cached features and scorer forward passes. L1 and calibratable L2 targets add a small label set without rebuilding a target-specific gradient datastore. The evaluation next tests whether this reuse preserves LESS-level quality, reduces registry cost, and transfers across support levels.

5 Experiments

The evaluation tests three claims: PCU-Select matches per-PEFT LESS on downstream quality, reduces the marginal cost of adding a PEFT, and transfers to new configurations according to their structural support in the training registry.

Setup

We evaluate on GSM8K, HumanEval, MMLU, and TyDiQA (Cobbe et al. 2021; Chen et al. 2021; Hendrycks et al. 2021; Clark et al. 2020), using exact match, Pass@1, accuracy, and F1, respectively. We express these scores as percentages before averaging. From a pool of 300K mixed instruction examples, every selector chooses 10%.

The main comparison covers five configurations seen during scorer training: attention and MLP LoRA, (IA)³, and adapters. The transfer study holds out configurations that vary rank, placement, learning rate, or family. Appendix Table 5 lists the full 17-configuration registry.

The four baselines differ in target specificity. **Random** uses neither task nor PEFT information. **RDS+** ranks frozen semantic embeddings by similarity to the task-sketch mean, but ignores the PEFT configuration. **Influence** ranks projected base-model gradients from one shared, PEFT-agnostic datastore. **LESS**, our per-PEFT implementation of gradient-influence selection (Xia et al. 2024), warms up each target adapter, builds an Adam-preconditioned gradient datastore, and ranks examples by sketch influence. PCU-Select uses its conditional scorer, uncertainty penalty, and adaptive semantic-cluster quotas. Appendix A gives full specifications.

We hold the backbone, PEFT recipe, training steps, and evaluation protocol constant, so only the selected subset changes. The main table varies target-training seeds 0, 1, and 2 while fixing the selector, scorer, sketch, and pool at seed 0. Matched seeds give every selector the same training randomness. We obtain each descriptive 95% interval by resampling the evaluated PEFT-task cells under this fixed selection pipeline.

PCU-Select Matches Per-PEFT LESS

Table 1 shows that PCU-Select improves on the PEFT-agnostic baselines and remains close to target-specific LESS. It gains 0.74 points over RDS+ and wins 17 of 20 PEFT×task cells, with descriptive interval $[+0.38, +1.06]$. It gains 0.50 points over Influence and wins 18 cells, with interval $[+0.21, +0.75]$. Its comparison with LESS is closer: a 0.04-point gap, interval $[-0.26, +0.33]$, and 10 wins.

The LESS comparison varies across tasks. PCU-Select gains on MMLU, stays within seed dispersion on GSM8K and HumanEval, and trails on TyDiQA and MLP-only LoRA. Task-grouped resampling yields $[-0.15, +0.22]$ for the aggregate gap (Appendix Table 3). **Takeaway.** Shared PEFT-conditioned supervision improves on PEFT-agnostic selection while preserving per-PEFT LESS’s average quality on the evaluated registry.

Amortization Cuts Marginal Cost by 44×

Table 2 separates *fixed/shared* cost, paid once, from *marginal* cost, paid for each new PEFT. All selector costs exclude the